

- 6 (a) Outline the role of DNA polymerase in the replication of DNA.

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..... [3]

- (b) Floxuridine is a drug used to treat cancer.

- (i) Floxuridine inhibits DNA replication.

Suggest why the inhibition of DNA replication is an effective treatment for cancer.

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..... [1]

- (ii) Floxuridine prevents thymine nucleotides being produced.

Explain why floxuridine does **not** affect transcription.

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..... [1]

- (c) Fig. 6.1 is a scanning electron micrograph of a lung cancer cell in a stage of the cell cycle.

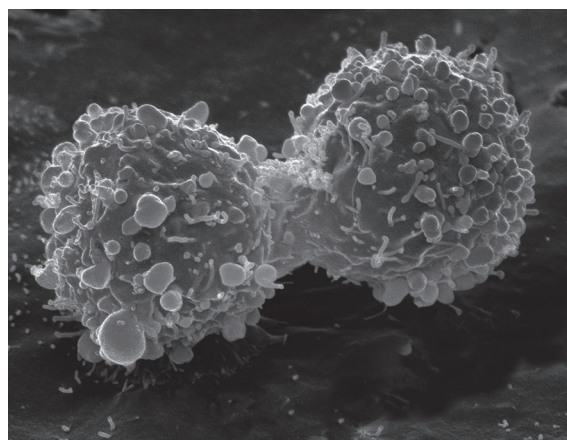


Fig. 6.1

- (i) Name the stage of the cell cycle occurring in Fig. 6.1.

..... [1]

(ii) Explain why smoking tobacco increases the risk of developing lung cancer.

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..... [3]

[Total: 9]

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